USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 21 LAB - Antenna Pattern
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

This packet is the L21 Lab, in three parts, and you turn it in once, complete, before Lesson 22. **Part 1, Predict**, is yours to do before the lab period: the shaded Predicted cells of each table, what your kit should read. **Part 2, Measure**, is your own kit's readings in the white Measured cells of the same tables, recorded in class. **Part 3, Analyze**, is yours to write after the lab.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt. I grade each Part 3 answer on four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before the lab period)

1. **Predict.** Fill in every shaded **Predicted** cell of the Setup table and Tables A to E. Work at the HB100 frequency your kit reported with **Find HB100** in the L18 Lab. If you did not record one, use the nominal 10.525 GHz and say so. Lesson 21's Pre-Lab frame names the frame that works each method, at 10.0 GHz. Attach your work for one row of each table on your own paper.

    - **The array.** $N = 8$ patches, E1 to E8, on a $d = 14$ mm pitch. Facing the patches, E1 is on the right and E8 on the left, and $+\theta$ is on the E8 side. The HB100 stands 1.2 m from the array.
    - **The sweep by hand.** It carries the HB100 from -90° to $+90^\circ$ along an arc of that radius in about 22 s, and your radius drifts by up to 10 cm either way.
    - **The update rate.** The slowest update rate is the one that still puts five readings on a first sidelobe during that sweep.
    - **Levels.** Report them as the tables do: Tables B and C in dBc, and Tables D and E relative to the Table D reference, the level with the kit at 0° and the beam at 0°.

Part 2: Measure (your kit, in class)

Follow the procedure on the L21 Lab page and fill in every white **Measured** cell. Before you record anything, set **Rx Gain** so that the main-lobe peak, with the source at boresight and the beam at $0°$, reads -10 ± 3 dBFS on the FFT, and leave it there for the whole lab. Keep the HB100 at patch-row height, 1.2 m from the array. Record each reading as you take it.

Setup	**Predicted**	**Measured**
Frequency (Signal Freq)		
Far-field distance / source distance		
Updates/s: slowest / on the plot title		
Rx Gain (dB)	–	

2. **Table A: check the angle axis.** With the beam sweeping as in the L18 Lab, turn the kit so the source sits at each angle below, and record the **Angle** readout, set to **Peak**. Turning the kit's face toward E1, as seen from the source, puts the source on the $+$ side, and the stand's scale reads the opposite sign: -30 on the stand puts the source at $+30°$.

Source angle (stand reads)	$-30°$ ($+30$)	$0°$ (0)	$+30°$ (-30)
Predicted Angle (Peak)			
Measured Angle (Peak)			

3. **Table B: the beam sweep.** Kit back at $0°$, beam sweeping. Read the trace.

Peak (dBFS)	**HPBW**		**Left sidelobe (dBc)**		**Right sidelobe (dBc)**	
Measured	**Predicted**	**Measured**	**Predicted**	**Measured**	**Predicted**	**Measured**

4. **Table C: the azimuth sweep by hand.** Beam held at $0°$. Carry the HB100 from $-90°$ to $+90°$ on the taped 1.2 m arc, aimed at the array, three times. Record each sweep's main-lobe peak and both first sidelobes.

	Sweep time (s)	**Peak (dBFS)**	**Left SL (dBc)**	**Right SL (dBc)**
Predicted	–	–		
Sweep 1				
Sweep 2				
Sweep 3				
Predicted spread	–		–	–
Spread (max $-$ min)				

5. Table D: the pass on the stand, beam held at $0°$. HB100 on its tripod at boresight, 1.2 m out, beam sweeping. One pass fills Tables D and E: each sweep sample is the beam held at that steer angle, so the trace at $0°$ steer is this table and the sample nearest $30°$ is Table E. Turn the kit to each mark of the stand and record the trace at $0°$ relative to its value with the kit at $0°$, in dB, at the source angle: the stand reading with its sign flipped, as in Table A. Write "floor" where the reading stops changing as you turn.

θ_{src}	$-90°$	$-80°$	$-70°$	$-60°$	$-50°$	$-40°$	$-30°$	$-20°$	$-10°$	$0°$
Level										

θ_{src}	$+10°$	$+20°$	$+30°$	$+40°$	$+50°$	$+60°$	$+70°$	$+80°$	$+90°$	
Level										

Reference: kit at $0°$, trace at $0°$ steer (dBFS) = ____

Now turn the kit slowly through each lobe and null below, and record the angle and level of its extreme, the highest reading for a lobe and the lowest for a null. Record every level relative to the reference above, as in the marks; Part 3 converts the lobes to dBc. Last, turn back to the main lobe and the right first sidelobe and read each again: the change is the stand's repeatability.

	Source angle		**Level**	
	Predicted	**Measured**	**Predicted**	**Measured**
Main lobe				
Left first null				
Right first null				
Left first sidelobe				
Right first sidelobe				
Main lobe, read again	–	–	–	
Right first sidelobe, read again	–	–	–	

6. **Table E: the pass on the stand, beam held at $30°$.** The same pass, read at the sweep sample nearest $30°$. Record every level, in the marks and in the lobe table, relative to the Table D reference.

θ_{src}	$-90°$	$-80°$	$-70°$	$-60°$	$-50°$	$-40°$	$-30°$	$-20°$	$-10°$	$0°$
Level										

θ_{src}	$+10°$	$+20°$	$+30°$	$+40°$	$+50°$	$+60°$	$+70°$	$+80°$	$+90°$	
Level										

	Source angle		**Level**	
	Predicted	**Measured**	**Predicted**	**Measured**
Main lobe				
Inner first null				
Outer first null				
Inner first sidelobe				
Outer first sidelobe				

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions. Convert the lobe levels of Tables D and E to dBc first: subtract that table's main-lobe level.

7. **Broadside, three ways.** Compare your beam sweep (Table B), your three sweeps by hand (Table C), and your measurement on the stand at $0°$ (Table D) with your predictions. Which features do the three methods agree on, and which do they not? Quantify the sources of error and explain any disagreement with your predictions.

8. **The beam at $30°$.** Compare your measurement at $30°$ (Table E) with your measurement at broadside (Table D) and with your predictions. What changed when the beam steered, and what did not? Explain each change, and describe a measurement on this kit that would test your explanation.

9. **By hand against the stand.** You measured the same held beam by carrying the source and by turning the kit. From the spread of your sweeps in Table C and your repeated readings in Table D, decide which quantities you would report from each method, and why. Then state this array's broadside first-sidelobe level as you would report it from today's data, with its uncertainty, and say what it doesn't explain.

Documentation: _____